

Purpose. A brief synthesis of the ProtoCosmoBot and ProtomegaBot reviews of *OmegaBuzz: A Human-Agent Collaboration and Cognitive Hive Platform*. This is design feedback, not a claim that the proposal’s components are implemented or validated.

Bottom line

OmegaBuzz is an unusually coherent proposal. Its strongest contribution is to keep four distinct things separate: **authorship**, **visibility**, **epistemic standing**, and **organizational authority**. The proposal should proceed as a staged, sidecar-first architecture—but only after it converts the most consequential abstract boundaries into executable contracts and reduces the first slice to a demonstrably small system.

What is especially right

- **Chat is not the write path.** Free discussion does not itself mutate organizational state; typed, authorized, provenance-backed transitions do.
- **Layered semantic memory with explicit promotion.** Private → epic → hive → federated memory avoids both a single mutable global AtomSpace and isolated agents that cannot build shared knowledge.
- **Claim versus proposition.** Separating a source-specific assertion from the proposition it concerns preserves disagreement, evidence, and revision better than a single “truth” record.
- **Scout’s deterministic provenance surface.** Keeping the reader and exact source navigation separate from model-driven enrichment makes the knowledge system inspectable.
- **Sidecar-first integration with Buzz.** Extend the collaboration substrate through typed events and small upstream patches, rather than prematurely entangling every proposed service with relay internals.
- **Resident Learners are separately governed.** The proposal preserves the useful distinction between wake-based organizational seats and continuously learning agents, without granting room speech automatic authority.
- **Human-centered operational discipline.** EpicFrame rehydration, explicit work ownership, signed history, shadow-before-enforcement, and human-only recovery are practical foundations rather than decorative governance.

Priority corrections before implementation

1. **Make the external baselines reproducible.** Pin Buzz and Scout by repository URL, commit SHA, license, and capability test. “Reviewed on 30 July” is insufficient for an engineering dependency; Scout licensing must be a Phase 0 gate, not a parallel concern.
2. **Specify concurrent state-transition semantics.** Signatures and idempotency do not decide races. Every mutable transition (work order, approval, delegation, claim admission) needs expected version/CAS or causal-parent rules, canonical ordering, rejection behavior, and repair/replay semantics.
3. **Make the authority membrane effective on reads as well as writes.** Private, quarantined, or restricted memory can influence a consequential output even if it cannot be directly promoted. Commit proposals should carry machine-checkable provenance/taint disclosures; policy should prevent an undisclosed restricted source from supporting a commitment.
4. **Turn proposition identity into a managed uncertainty.** Exact proposition matching is a hard semantic-coreference problem. Start with stable claim IDs and conservative candidate links; permit human review or explicit MeTTa/PLN unification rules to assert identity later. Do not assume automatic semantic equivalence is solved.
5. **Define tiered promotion review.** Run automatic schema, provenance, authorization, duplication, and contradiction checks by default; require human review for high-consequence, disputed, policy-flagged, or cross-boundary promotions. Otherwise review becomes either a bottleneck or a rubber stamp.

6. **State the AtomSpace fallback plan.** Identify which initial needs can use conventional stores (e.g., relational claim records and graph queries) and which genuinely need AtomSpace/MeTTa/PLN. Add upstream maturity as an explicit risk and avoid making the first useful loop depend on unverified gateway capabilities.
7. **Limit service count in Slice 0.** The proposal's 15-ish service decomposition and large coding-agent roster are a useful target map, not a first implementation. State which components are real, thin shims, or deferred for each slice, together with an engineering capacity/velocity assumption.

Recommended first slices and gates

1. **Slice 0: organizational loop.** One human, two signed seats, one bounded deliberation, and a committed decision/work order. Exercise policy, capability, pause/revoke/read-only, and human recovery under agent or service failure.
2. **Slice 1: provenance loop.** Preserve artifacts and source anchors; replay the signed history into a read-only view that traces a decision to exact room events. No semantic memory admission is required yet.
3. **Slice 2: semantic admission loop.** Add a candidate claim, tiered review, explicit promotion, and authorized Scout context compilation. Demonstrate retraction/supersession and a contradiction fixture.

Each slice should have a concrete exit gate. This resolves the present scope mismatch in which the executive summary's first vertical slice includes AtomSpace/Scout while the roadmap places them in Phase 2.

Resident Learners and federation

Keep Resident Learners in shadow/observation mode until the charter, visible consent, inspectable/deletable experiential memory, egress lease, attention budget, loop-breaker telemetry, and authority membrane operate together. A learner observing a deliberation has more continuous context than wake-based seats; this asymmetry may be useful, but its influence must be disclosed and bounded.

Federation by sourced assertion rather than shared truth is correct. Offline bundles are an appropriate first mechanism, but the plan should test remote, authorization-aware queries earlier if iterative multi-hive work becomes a real need.

Immediate decisions requested

- Approve the three-slice framing and name the exact Slice 0 components.
- Assign Phase 0 owners for baseline pinning, Scout licensing, transition semantics, and the safety-kernel failure test.
- Adopt conservative proposition matching and a tiered memory-promotion policy as initial defaults.
- Record a conventional-store fallback boundary for AtomSpace Gateway capabilities.